Question 1

A rectangular region of land is divided into 54 square lots of equal area S in square units. The length of the region is 1.5 times the width of the region. If the width of the region is $a\sqrt{S}$ units, what is the value of a?

A. 5

B. 6

C. 7

D. 8

Answer to Question 1

Correct Answer: B

Question 2

In the given pair of equations, a and b are constants. The graph of this pair of equations in the xy-plane is a pair of perpendicular lines. Which of the following pairs of equations also represents a pair of perpendicular lines?

$$5x + 7y = 1$$
$$ax + by = 1$$

A. $10x + 7y = 1, \quad ax - 2by = 1$

B. $10x + 7y = 1, \quad ax + 2by = 1$

C. $10x + 7y = 1, \quad 2ax + by = 1$

D. $5x - 7y = 1, \quad ax + by = 1$

Answer to Question 2

Correct Answer: B

Question 3

A real estate company offers a series of three webinars. 2,000 people attended the first webinar. 65% of the people who attended the first webinar attended the second webinar, and 44% of the people who attended both the first and second webinars attended the third webinar.
If those who attended the first but did not attend the second webinar, 31% attended the third webinar. How many people attended both the first and third webinars but did not attend the second webinar?

Answer to Question 3

Correct Answer: 217

Question 4

The functions f and g are defined by the given equations, where $x \geq 0$.

$$f(x) = 19(1.27)^x + 43$$
$$g(x) = 8(0.77)^x$$

Which of the following equations displays, as a constant or coefficient, the maximum value of the function it defines, where $x \geq 0$?

A. I only

B. II only

C. I and II

D. Neither I nor II

Answer to Question 4

Correct Answer: B

Question 5

Circle F in the xy-plane is represented by the equation:

$$(x-13)^2+(y-9)^2=196$$

Circle G is obtained by shifting circle F 5 units to the left and 11 units up.
An equation representing circle G is:

$$(x+h)^2+(y+k)^2=196$$

where h and k are constants.
What is the value of $h+k$?

Answer to Question 5

Correct Answer: -28

Question 6

The function q is defined by:

$$q(x)=a|x-9|^2-87|x-9|+b$$

where a and b are constants, and $a>b>87$.
If $q(639)=h$ and $q(-621)=k$, where h and k are constants, what is the value of:

$$9(-639)^{h-k}+621(9)^{k-h}?$$

Answer to Question 6

Correct Answer: 630

Question 7

In the given equation, a and b are positive constants. A solution to the equation is $x = 42$.

$$x(9x - 2a) + b(9x - 2a) - (x + b) = 0$$

What is the value of a?

Answer to Question 7

Correct Answer: 188.5

Question 8

A right rectangular prism has a base area of $28p$ square centimeters (cm^2). The length of the base of the rectangular prism is 12 cm, and the height of the rectangular prism is 4 cm.
Which expression represents the surface area, in cm^2, of the right rectangular prism?

A. $\frac{512p}{3}$

B. $\frac{224p}{3} + 96$

C. $56p + 192$

D. $560p + 96$

Answer to Question 8

Correct Answer: B

Question 9

The expression

$$\frac{x^{-2}y^{\frac{1}{2}}}{x^{\frac{1}{3}}y^{-1}}$$

where $x > 1$ and $y > 1$, is equivalent to which of the following?

A. $\frac{\sqrt{y}}{\sqrt[3]{x^2}}$

B. $\frac{y\sqrt{y}}{\sqrt[3]{x^2}}$

C. $\frac{y\sqrt{y}}{x\sqrt{x}}$

D. $\frac{y\sqrt{y}}{x^2\sqrt[3]{x}}$

Answer to Question 9

Correct Answer: D

Question 10

The function z is defined by the equation:

$$z(w) = (0.829)^{2w}$$

The value of $z(w)$ decreases by $p\%$ for each increase by 1 in the value of w. Which of the following is closest to the value of p?

A. 0.171

B. 0.829

C. 17.1

D. 31.3

Answer to Question 10

Correct Answer: D

Question 11

$$-16(5x-3)^2 + 4(5x-2)^2$$

The given expression can be rewritten as:

$$\frac{a}{4}x^2 + \frac{b}{4}x + \frac{c}{4}$$

where a, b, c are constants. What is the value of $a + b + c$?

A. -112

B. -56

C. -28

D. -7

Answer to Question 11

Correct Answer: A

Question 12

The graph of line h is shown in the xy-plane. Line k (not shown) is defined by the equation:

$$sx + 40y = t$$

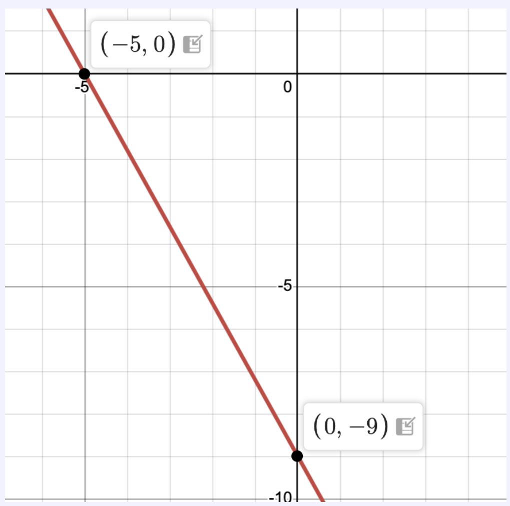
where s and t are constants. If line k is graphed in this xy-plane, the result is a graph of two linear equations. This system of two linear equations has no solution.
Which of the following is **NOT** a possible value of t?

A. -360

B. -9

C. 72

D. 200



Answer to Question 12

Correct Answer: A